

Micromouse Final Project — Bill of Materials

Component	Store Link
VL53L0X ToF Distance Sensor (x3)	https://microhm-eg.com/ev-53-v5330x-laser-tof-flight-time-ranging-sensor-module-high-precision-time-of-flight-distance-measurement-with-pwm-serial-output/
MPU6050 IMU — GY-521 Breakout (x1)	https://makerselectronics.com/product/mpu-6050-imu-sensor-module-with-6-dof-gyroscope-accelerometer/?srsltid=AfmBOqgFhRUrtwbtsu45byAbVxK5yC81CpLpin_ahIZtsCqxIXO
OLED Display 0.96" SSD1306 (x1)	https://makerselectronics.com/product/oled-display-12c-4-pin-0-96x0-48x0-1-3-2v3id=AfmBOqo9TzWd4RaoD8UfQAHxTZu8DELLkChVulD_OuO_ICUz2WC8-G
Resistor 2.2kΩ, SMD 0805 (x2)	https://makerselectronics.com/product/chip-resistor-smd-2-2k5cF989-%29b11-0-125w-%29b1100ppm-%2%84%83-0805/
Resistor 100kΩ, SMD 0805 (x1)	https://makerselectronics.com/product/chip-resistor-smd-100k5cF989-%29b11-0-125w-%29b1100ppm-%2%84%83-0805/
Resistor 33kΩ, SMD 0805 (x1)	https://makerselectronics.com/product/chip-resistor-smd-33k5cF989-%29b11-0-125w-%29b1100ppm-%2%84%83-0805/
Resistor 10kΩ, SMD 0805 (x1)	https://makerselectronics.com/product/chip-resistor-smd-10k5cF989-%29b11-0-125w-%29b1100ppm-%2%84%83-0805/?srsltid=AfmBOqgyUDhm6byjnpKN3UK7W3uDSXU-A_i_v2PsaD8BKzVp1KY6
6-pin JST Connector, JST-PH 2.0mm (x4)	https://makerselectronics.com/product/66b-xh-6-jst-xh-6-pin-header-2-5mm/
M3 Standoff / Spacer (x4)	https://www.ram-e-shop.com/en/shop/spacer-copper-15mm-copper-pcb-spacer-15mm-m-f-6291?category=121
M3 Screw + Nut (x4)	https://www.ram-e-shop.com/en/shop/spacer-copper-15mm-copper-pcb-spacer-15mm-m-f-6291?category=121
2-pin JST Connector, JST-PH 2.0mm (x1)	https://makerselectronics.com/product/b2b-xh-6-jst-xh-2-pin-header-2-5mm/
Fuse — PTC Resettable, SMD 1206 (x1)	https://makerselectronics.com/product/resettable-fuses-ptc-6v-1-1a-wt-1206-110-smd/?srsltid=AfmBOoqVqYQzveVAIXUjTXOMKfY2efAu5du8C7Bgh7uAgGooB85E
Switch — SPDT Slide Switch (x1)	https://makerselectronics.com/product/slide-switch-3-pin-on-off-spdt/
MOSFET — IRF9540, P-channel, TO-220 (x1)	https://makerselectronics.com/product/irf9540-p-channel-power-mosfet-transistor-to-220-100v-19a-0-20%CP98/?srsltid=AfmBOoq1jdt_8fLwHtT3yw9ubHnZzph1eqs7G8pt_mWhbqWddqVeb
Voltage Regulator — AMS1117-5.0, SOT-223 (x1)	https://makerselectronics.com/product/jootwork-ams1117-5-70db120hz-1a-fixed-5v-positive-18v-sot-223-dropout-regulators-smd/?srsltid=AfmBOoq1oD1Co0kV48R2HTIM8japGcm8t5o_vlV1DW18RnXideLU0eQ
Capacitor 10µF, SMD 0805 (x4)	https://age-one.com/product/smd-chip-ceramic-capacitor-size-0805-10uf/?srsltid=AfmBOoqWTPYNBSQLMGHv8kxTkc_UYvH3HYpkiimbdf1pGvF5TmQoqX
Capacitor 0.1µF, SMD 0805 (x6)	https://age-one.com/product/smd-chip-ceramic-capacitor-size-0805-100nf/?srsltid=AfmBOoqVe7Hw8_FNAFO4PG_d4yx-5z71rQnOW0bwQAOZ3SM_-Ubm9VG
Capacitor 2.2µF (6.3V), SMD (x2)	https://makerselectronics.com/product/multilayer-ceramic-capacitors-mlcc-smd-smt-2-2uf-%29b110-50v-0805/
Capacitor 0.01µF / 10nF (16V), SMD (x2)	https://age-one.com/product/smd-chip-ceramic-capacitor-size-0805-10nf/?srsltid=AfmBOoqgTvy194PIW-PSQYVBentdDIZ-Hh_RipjFAhIedq8Nzov
ESP32-C3 Super Mini (x1)	https://makerselectronics.com/product/esq32-c3-super-mini-development/?srsltid=AfmBOoqXv7Zw6v66kV6dHSW47cQMP8Dn8t5oWpaD4uQnFjv48983G
Motor Driver IC — DRV8833PWP, HTSSOP-16 (x1)	https://makerselectronics.com/product/drv8833-2-channel-dc-motor-driver/?srsltid=AfmBOo86k1wglbVtMfgz31TrRhgCuAlpGvVT77z2eQdL-G68C7ZF/
N20 Micro Gear Motor w/ Encoder — JGA12-N20 (x2)	https://makerselectronics.com/product/jga12-n20-micro-metal-gear-dc-motor/?srsltid=AfmBOoq7n4NTrn94I4Z1mVXXMG1kHv6ioDZUuHElzzuTSmGT5l8Qwm
LiPo Battery Cell — ECOCELL 3.7V 1000mAh (x2)	https://store.fut-electronics.com/products/3-7v-rechargeable-polymer-lithium-ion-battery-ecocell/?srsltid=AfmBOoKnTVOS8nG7AhAcvC87nCP2p7ltnmniKWpD4QwFE-L48ZTE
Metal Castor Wheel (x1)	https://age-one.com/product/caster-wheel-small/?gad_source=1&gad_campaignid=24034682861&gbraid=5AAAAAGAZtmeQ1CauZTUaF-WXGawwF7I8&clid=CwKCAJwvwyTBhBaEfwAmf-3nurhta6SHFLLY6Ubb0vhadidQIP7Kgn1JWfflxXbUstV83BplUzoChwQAvD_8fwf
M2 Screw (x8)	https://microhm-eg.com/screw-m2-2mm-x-12mm-with-nut/
M2 Nuts (x8)	https://microhm-eg.com/screw-nut-m2-2mm-1pcs/
N20 Motor Mount Bracket	https://makerselectronics.com/product/n20-motor-mount-bracket/?srsltid=AfmBOqgb89J5QH_-bfHGuDRGu9apPYB8EPKv84FTWnkwQVGus
N20 Rubber wheel	https://microhm-eg.com/n20-48mm-minio-rubber-wheel-for-n20-gear-motor/
Capacitor 100µF (6.3V), SMD (x1)	https://makerselectronics.com/product/multilayer-ceramic-capacitors-mlcc-smd-smt-6-3v-100uf-x-5r-%29b120-1206/?campaignid=20503411856&adgclid=CwKCAJwvwyTBhBaEfwAmf-3nurhta6SHFLLY6Ubb0vhadidQIP7Kgn1JWfflxXbUstV83BplUzoChwQAvD_8fwf